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Methodology for the design and development of serious games as a tool for the treatment of ADHD-related symptoms in children

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ABSTRACT

In recent years, serious video games have emerged as effective tools to support the educational and therapeutic needs of children, proving to be intellectually stimulating and successful in achieving specific objectives. This article focuses on the methodology for designing and developing serious games tailored to address deficits in children diagnosed with Attention Deficit Hyperactivity Disorder (ADHD). To ensure the quality and effectiveness of the games, the proposed methodology integrates the findings from a comprehensive review of the key characteristics that video games should meet in order to satisfy the therapeutic and cognitive needs of children with ADHD. As part of the video games co-design process, a multidisciplinary team was formed, comprising professionals from psychology, education, and computer science. Additionally, the methodology includes an evaluation phase to validate the games' playability, engagement, and adherence to the established design criteria. The sample consisted of 12 adult participants (6 males, 6 females) and 13 neurotypical children, aged 8 to 12 (6 girls, 7 boys). The three games developed were implemented on mobile devices to guarantee accessibility and portability in therapeutic and educational settings. Preliminary results confirm the successful inclusion of key elements that video games should feature to address the specific needs of ADHD children, as well as the positive evaluation of the games in terms of Playability and Sympathy Enjoyment. These findings highlight the potential of a structured design methodology as a replicable model for developing therapeutic interventions for children with this disorder using serious games.

1. Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental disorder typically diagnosed in childhood, affecting approximately 3-5% of school-aged children worldwide [1]. Given its prevalence, ADHD has become one of the most extensively studied subjects in this population. According to Brito et al. [2], ADHD can present with predominant symptoms of inattention, hyperactivity/impulsivity, or a combination of both, leading to disruptions in behavioral, cognitive, academic, and social functioning. Individuals with ADHD frequently experience difficulties in executive functioning, including working memory, inhibitory control, and self-control, as well as challenges related to motivational style and sensitivity to rewards and punishments. These difficulties manifest in an inability to inhibit behavior, plan and organize responses, and regulate effort [3]. Additionally, children with ADHD also exhibit deficits in motor skills, particularly in visuomotor coordination, which affects their ability to perform tasks requiring motor precision. Specifically, Goulardins et al. [4] have indicated that improving these motor skills not only facilitates the development of motor abilities but is also associated with improvements in attention and inhibitory control, two of the key cognitive deficits in ADHD. Addressing these deficits, particularly through interventions targeting, among others, visuomotor coordination, could significantly contribute to the comprehensive management of ADHD symptoms. Effective instructional strategies have been demonstrated to foster an environment in which children with ADHD can learn more effectively. Among these are digital interventions, which provide various treatment options and educational benefits for children with ADHD [1]. These interventions are highly accessible and can be implemented in various settings, such as homes, schools, and clinics [5].

Digital tools enable self-application and customization, offering flexibility while preserving privacy and anonymity. Children and

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adolescents, as avid consumers of digital technology for social and recreational activities, are often enthusiastic participants in digital interventions [6]. One of the most widespread types of digital applications today is video games, which can transport users to non-real worlds and allow them to interact with these imaginary environments through multiple sensory channels, such as visual, auditory, kinesthetic and tactile [7]. Within these, serious games (SGs) are distinguished by their focus on objectives beyond entertainment, though this does not imply they lack engagement [8,9]. For this reason, SGs have gained popularity in fields outside of entertainment. SGs 49320462699have emerged as valuable tools in education, providing dynamic and playful resources and motivational intervention strategies that diverge from conventional therapies [10–12]. Furthermore, SGs show promising applications in neurodevelopmental disorders, offering novel ways to support learning, social interaction, and the inclusion of students with specific educational support needs [13,14]. In both cases, provided that the objectives to be achieved, the content to be addressed, the evaluation procedure and the skills to be developed are clearly defined [5].

Intervening with children diagnosed with ADHD is a complex task due to the individual differences in the manifestation of the disorder [15]. Currently, several interventions are available to improve ADHD symptoms, including pharmacological, behavioral, and cognitive approaches. Authors such as Jarque Fernández [16], Sánchez [17] propose that these interventions should not be implemented in isolation but as a multicomponent strategy to achieve the most optimal and rigorous outcomes. Taking into account that children are now considered digital natives, Camacho-Conde and Climent [18], Delgado-Reyes and López [19] advocate for the use of SGs as complementary intervention tools to enhance ADHD symptoms in children, since relying solely on them may not be enough [20]. Overall, SGs serve as an excellent tool to enhance attention, effort, and motivation while developing competences and skills such as mental agility, promote comprehension, reflection, and strategic reasoning [21].

Although studies on SGs have demonstrated their effectiveness in various neurodevelopmental disorders and specific learning difficulties, research specifically exploring their impact on ADHD symptoms remains limited. These findings underscore the need for further investigation to examine the potential benefits of video games as intervention tools for ADHD. In addition to this, it should be added that there remains a gap in the development of standardized methodologies for creating ADHD-specific serious games. This study addresses this gap by presenting a replicable framework that bridges therapeutic goals with engaging game design. This paper focuses on presenting a methodological framework for the design, development, and evaluation of SGs specifically tailored for children with ADHD, implemented on mobile devices to guarantee accessibility, portability, and ease of use in therapeutic and educational settings. The methodology emphasizes integrating evidence-based design principles to ensure therapeutic and cognitive effectiveness. This structured approach aims to create tools that not only engage children but also improve key skills such as attention, inhibitory control, and visuomotor coordination. By documenting this framework, we aim to provide a replicable model that can guide future developments in the field of educational and therapeutic games. The three serious games reported in this study are presented as functional prototypes created to validate the proposed methodological framework prior to clinical implementation.

Following this methodology, a set of games specifically designed for children aged 8 to 12 years with ADHD was developed. To ensure the quality and effectiveness of these games, the process began with a design analysis phase that included a comprehensive literature review to establish ADHD-specific video game design criteria. Based on these criteria, the design and development phases were carried out, focusing on incorporating key aspects such as motivational elements, sensory control, and manageable difficulty. Accordingly, the specific aim of this work was to evaluate the playability and engagement of the developed games, as well as to assess their adherence to the established design criteria and their performance regarding the quality factor of accuracy percentage. These aims are reflected in the following two hypotheses:

H1. Video games developed based on specific criteria for children with ADHD exhibit high levels of playability, engagement, usability, and accuracy across expert and non-expert users.

H2. The key design aspects incorporated, such as motivational elements, sensory control, and manageable difficulty, ensure the games' adaptation to the specific needs of children with ADHD.

Additionally, as a positive reinforcement tool to enhance the child's engagement in the intervention [22–24], an optional augmented reality (AR) game was included to enable real-time interaction and foster creativity.

It should be noted that the purpose of this work is not to provide a software toolkit, but rather a conceptual and methodological framework that guides the design, development, and evaluation of serious games for children with ADHD. The three mini-games included in this study serve as illustrative examples of how this framework can be applied in practice.

Next, Section 2 (Related Work) provides an overview of previous studies and evidence on the use of serious games for ADHD, serving as the background to frame our proposal. Section 3 (Methodological Framework for the Creation of SGs for Children with ADHD) then details the analysis, design, and development phases of the games. Section 4 (Evaluation) outlines the evaluation process, while Section 5 (Results) presents the findings. Section 6 (Discussion) interprets the results and offers suggestions for future work. Finally, Section 7 (Limitations and Future Work) provides the final remarks of the study.

2. Related work

There is extensive literature that demonstrates the effectiveness of the use of SGs in a variety of representative symptoms of this disorder in children with ADHD. Specifically, SGs provide an opportunity to enhance both cognitive and behavioral abilities in this population [15,25]. They can enhance motivation, skill development, symptom management strategies [26], as well as increase interest in and adherence to therapeutic interventions [27,28].

According to Bioulac et al. [29], the use of SGs improves visual-motor coordination, spatial attention, executive functions and inhibitory control, as well as stimulate motivation, effort and arousal to a greater extent than conventional therapies. In their studies, García-Redondo et al. [5], Muñoz et al. [30] asserted that SGs can address waiting, planning, instruction-following and goal-achievement abilities. Other research like in Bediou et al. [31], suggests that SGs can significantly enhance attention and spatial cognition. Moreover, SGs have shown promise in improving time management, planning skills, and reducing hyperactivity symptoms in students with ADHD, as demonstrated by Bul et al. [32], Lumsden et al. [33].

To improve attention in the infant-juvenile period in ADHD children, Bahana et al. [34] developed a video game specifically designed to target memory and visuospatial tasks in children between the ages of 7 and 12. Furthermore, García-Redondo et al. [5] conducted a study involving SGs training in children with ADHD, demonstrating improvements in attentional performance, particularly in concentration and accuracy on tasks [35,36]. Some video games directly address attention without the need for a narrative component [37]. In this virtual environment, participants interact with a virtual teacher who guides them in working on different types of attention, including selective and sustained attention. Recent works have also demonstrated that serious games can enhance attentional and executive functions in neurodiverse children, including those with ADHD [38].

In 2021, Guerrero García and González Calleros [15] developed seven educational video games for children with ADHD. Their findings revealed improved cognitive performance, increased motivation among participants, and enhanced inhibitory and attentional responses. Regarding the improvement of inhibitory control and psychomotor skills, a video game "*Mental Brain*" was found to have a positive impact on the cognitive flexibility of children with ADHD, particularly in terms of inhibition and visuospatial functions [39]. The video game proposed by Benzing and Schmidt [40] showed benefits in improving motor skills through strength, coordination, and endurance training to enhance executive functions in children with ADHD.

In most video game-based therapies for children with ADHD, it is recommended that adults supervise the sessions to maximize benefits and ensure the safe and effective implementation of the treatment. Furthermore, various studies support the involvement of adults as a key factor in these interventions, as it contributes to both the development and validation of video games. These adults are typically individuals who either live with children with ADHD or work with them, such as teachers, therapists, or caregivers. This involvement allows for better adaptation of the video games to the specific needs of the disorder and ensures their effectiveness before they are implemented with children [41,42]. Initially testing games on adults increases the likelihood of developing high-quality interventions that are engaging, effective, and beneficial for children with ADHD. The mobile application proposed in Jácome et al. [41] aimed to make the prototype created as attractive as possible and improve divided attention in children with ADHD. To validate the prototype, they conducted an experiment with five children, their parents, and therapists. Parents provided feedback on the child's acceptance of the prototype and suggested improvements such as changing some of the interface images that they felt were not clear. The results showed a 100% acceptance rate of the application reported by the children, parents, and psychologists. Similarly, in Bul et al. [42], a SG was developed with the aim of promoting behavioral learning and encouraging the use of strategies in domains of daily life functioning of children with ADHD (time management, organization, and prosocial skills). The game was accepted by both ADHD children and their parents, and parents' opinions helped to make well-informed decisions about the frequency of children's play.

The works proposed by Sonne and Jensen [43], Bernardelli et al. [44] illustrate examples of games that were initially tested on adults before being used with children. In collaboration with ADHD professionals, Sonne and Jensen [43] developed a breathing-controlled video game. The goal was to keep the child's attention and increase relaxation by combining a breathing exercise with a video game [43,45]. The game was evaluated by 16 adults and was found to have a positive effect, assisting participants in achieving a state of relaxation similar to that achieved through traditional breathing exercises. A serious Virtual Reality (VR) game was presented in Bernardelli et al. [44]. This application aimed to create a meaningful and motivating environment for the child, capable of supporting the development of attentional components while providing an engaging experience. The development and validation process involved adult experts in both VR and bioengineering. The unanimous positive feedback received from the participants led to the conclusion that the application worked and was easy to use.

Recent studies have outlined several key principles for the design of serious games aimed at ADHD interventions. Sújar et al. [46], De Luca et al. [47] and Guerrero García and González Calleros [15] emphasize the importance of basing game design on user interests, incorporating cognitive exercises targeting attention and impulse control, and ensuring gradual difficulty progression. Furthermore, they recommend providing constant feedback through elements such as points, badges, and leaderboards to maintain user engagement. These principles were integrated into the design of the video games in this study, ensuring that the gameplay experience was both engaging and therapeutically beneficial. Following these recommendations, the games in this study were designed to promote these skills, with sessions structured to include task variability and short breaks (every 10 minutes) to optimize engagement and minimize fatigue. Furthermore, interventions such as Project Evo [48], developed by Akili Interactive Labs and approved by the U.S. Food and Drug Administration (FDA), share many of these design principles. This therapeutic video game is based on similar foundations to those implemented in our study, including progressive difficulty adjustment, continuous feedback, and a focus on impulse control and attentional processes. Like Project Evo, the games developed in this research were designed to promote key cognitive skills through structured mechanics and interactive tasks that can be adapted to each player's performance.

Compared to the games reported in previous studies, the serious games proposed in this work are distinctive in that they were designed within a structured methodological framework that explicitly considers the analysis, design, and development phases. While earlier approaches often focused on isolated cognitive skills such as attention, memory, or motor coordination, our games integrate multiple design principles relevant for ADHD: motivational reinforcement, the management of distractors, progressive levels of difficulty, and mechanisms for feedback and reward. Moreover, the present study not only describes the games themselves but also provides a replicable methodological process for their development, which is less common in prior literature. This methodological orientation, combined with preliminary evaluation with both neurotypical children and adults with ADHD, establishes a foundation for future clinical validation.

3. Methodological framework for the creation of serious games for children with ADHD

It is important to clarify that the contribution of this study is not the development of a software toolkit or technical platform, but rather a conceptual and methodological framework that guides the design, development, and evaluation of serious games for children with ADHD. This framework is structured into four main phases (analysis, design, development, and evaluation), each informed by evidence-based principles such as motivational reinforcement, management of distractors, progressive levels of difficulty, and the role of adult supervision. The three mini-games presented in this paper should be understood as illustrative examples of how the proposed methodology can be applied in practice, rather than as the final outcome of the work. In this way, the methodology offers a replicable process that other researchers and practitioners can follow to design and evaluate new games targeting ADHD.

The creation process of the SGs described in this study was based on a structured methodology consisting of four consecutive phases: Analysis, Design, Development, and Evaluation. The analysis phase focused on identifying and studying the specific requirements for designing games aimed at children with ADHD. Based on this foundation, the team proceeded with the design and development phases, incorporating key elements such as the game environment, mechanics to enhance targeted skills, interactive objects, reward systems, and technical specifications derived from the analysis. These phases ensured that the games are tailored to the specific needs of this population, following evidence-based design principles. Fig. 1 presents a horizontal flowchart that visually summarizes the structure of the methodological process, highlighting the main actions and tools involved in each phase.

3.1. Analysis phase

A multidisciplinary work group consisting of five specialists from three different academic areas (psychology, education and computer graphics) was formed. The team worked together with the main objective of identifying the key elements for the design of the games that addressed ADHD-related symptoms. Four meetings were conducted to establish the general characteristics of the battery of games to be developed. This phase established the foundation for creating games, ensuring that they were tailored to meet the specific needs of children with ADHD. Table 1 presents these characteristics.

49320444739During the initial meeting, target audience were identified, and therapists defined the skills they aimed to enhance by using video games as a complementary tool to conventional interventions. Subsequent meetings focused on reviewing existing

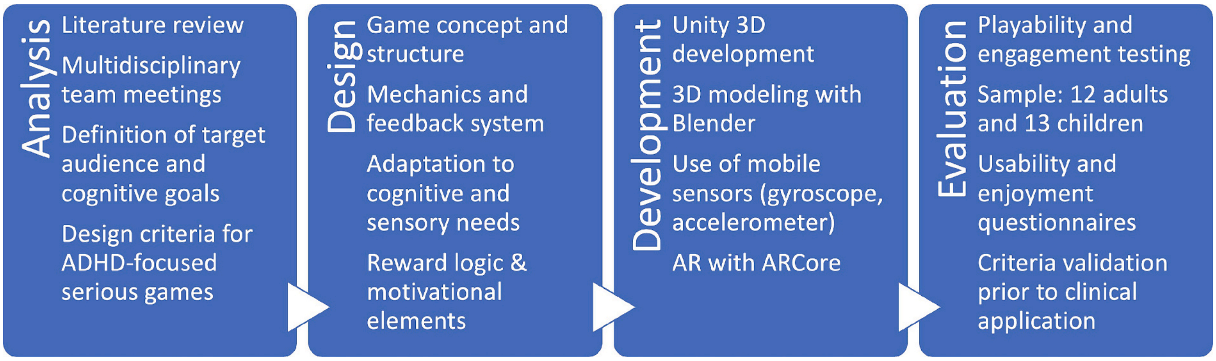

Fig. 1. Overview of the four-phase methodology used to design, develop, and evaluate serious games for ADHD children.

Table 1
Characteristics collected in the analysis phase. Supporting references are included to facilitate reading.

Target audience	Children aged 8-12 years with ADHD
Source: Literature & co-design team (psychologists/educators) [15,18] 49320442827	
Skills to be enhanced	Level of attention
Source: literature & therapeutic goals defined by psychologists [29,49]	Inhibitory control
Visuomotor coordination 49320442959	
Relevant agreed aspects to be implemented in the video games	Motivational character
Source: literature & multidisciplinary team (psychologists/	Playful activities
educators/computer scientists) [5,50]	Without excessive auditory and visual stimuli Degree of difficulty affordable
493204451926Estimated playing time Immediate evaluation Reward based system	
User interface	Intuitive and user-friendly
Source: based on best practices (developer team & educator input) [15] 49320442834	
Number of games to be developed	3 intervention games Source: multidisciplinary team, based on scope & therapeutic goals [41,42] 1 reward game
Game platform	Smartphone & tablet
Source: developer feasibility & team accessibility input [23,51] 49320442808	
Operating system	Android
Source: developer decision (devices & implementation constraints) [52]	

research and guidelines to establish a solid theoretical foundation for the development of the game suite, ensuring its design aligned with the research objectives.

The following summarizes the key aspects identified in the literature review and the proposed solutions for integrating these criteria into the games:

- **Motivational character**. According to Guerrero García and González Calleros [15], Echeverry Chaves [49], positive reinforcement plays a crucial role in enhancing motivation and awareness of progress in participants. Positive reinforcement should be audio-visual and emphasize achievements rather than mistakes to avoid frustration [15]. It was proposed that the game displayed only a counter of successes achieved, allowing the child to focus solely on their progress rather than on their mistakes. In addition, each time the child finishes a game, positive reinforcement was provided through auditory and visual messages such as "Wow" and "Congratulations" and the score achieved.
- **Playful activities**. The attention, coordination and planning tasks to be performed in the video games are approached as games to avoid monotony. In the proposed games, the interval between stimuli the child receives was not excessively long, so that performance is not affected [50].
- **Without excessive** audio **and visual stimuli**. Activities should support the child's attention and concentration with auditory-visual reinforcement. Extremely loud noises can cause auditory hypersensitivity and thus disturb and disrupt the concentration and motivation of children with ADHD [49]. It has been proposed that only sounds indicating success or failure should appear for each action performed. Furthermore, to enhance the motivational aspect, successes are accompanied by a higher musical tone than mistakes. During playtime and in the waiting periods between games, the music used will not be flashy or strident, and will not have lyrics. Pastel tones were avoided due to their low contrast and limited saliency [15,49]; instead, primary colors were used to improve visual clarity and minimize distraction. Additionally, the scenes will be as clean as possible, without any distracting elements. To minimize distraction, one of the video games features a simple avatar (a plane) without animations, as it serves solely as an element to be moved with the mobile phone. The other two games are in the first-person perspective, without any avatars.
- **Level of difficulty affordable**. Game instructions must be clear and concrete [49]. Each video game proposed included a demo with a video tutorial to show how to play as well as an option to practice the games before starting the actual gameplay. Furthermore, according to Guerrero García and González Calleros [15] the difficulty level must be manageable for the participants. In the study proposed by Bioulac et al. [29], it was determined the difficulty of each level based on the reduction of time to achieve the video game objectives or the increase of distracting stimuli. The proposed games had three levels of difficulty: easy level (L1), medium level (L2) and hard level (L3). Additionally, the LexiaReadable-Regular font will be used for the texts. This font is specifically designed for dyslexic individuals, as dyslexia is a common comorbidity in people with ADHD. It should be noted that in this initial version, difficulty adjustment is static and limited to these predefined levels (L1–L3), following the recommendations of Guerrero García and González Calleros [15] and Bioulac et al. [29].
- **The time spent playing**. The literature on video game intervention for children with ADHD emphasizes the significance of the time spent playing video games. Excessive game exposure to gaming can lead to irritability, anxiety, and prolonged periods of inattention [26]. In line with this, Chan and Rabinowitz [53] find that children with ADHD who play video games for more than one hour show greater levels of inattention compared to those who play for less than one hour, recommending to play for an average of 30 minutes [40,54]. It

has even suggested that the time spent playing serious video games should be reduced to 10 minutes [5]. The three video games that will be developed in this work had a duration of approximately 8–9 minutes each, including the preparation phase of the game.

- **Immediate evaluation.** Immediate feedback enables players to track their progress and analyze their mistakes. In the video games created, at the end of each level of difficulty, the achieved results will be displayed on the screen [5].
- **Reward system.** While children are playing the game and successfully achieving goals, a positive musical stimulus will play [15,55]. Upon completing the game and making satisfactory progress, the screen will display positive visual stimuli such as "*Congratulations!*" or "*Wow!*" accompanied by a musical stimulus to indicate successful completion of the game [15,49]. Points will be awarded for successes, while failures will not deduct points, in line with the recommendation to provide positive reinforcement and avoid frustration in children with ADHD [49,55]. After each game, the points earned will be displayed on the screen. In addition, at the end of each of the three games, the child will have the option to participate in a fun reward game designed to promote creativity and motivation, in line with previous work highlighting the importance of secondary reinforcements such as AR-based experiences [15,23,24].

After identifying key design and development characteristics for video games tailored to children with ADHD, the creation of three minigames is proposed, aimed at improving sustained and selective attention, inhibitory control, and visuomotor coordination. To further enhance engagement, a reward-based game was also developed, allowing children to express their creativity by drawing graffiti using only their mobile device.

3.2. Design and development phases

The following subsection describes the design and development of the three prototype games created according to the proposed methodological framework. The technical team responsible for creating the SGs took into consideration the recommendations, knowledge and resources suggested by the professionals involved, as well as the characteristics that make these games suitable for children with ADHD respecting the design principles for children's serious games [15,56] and following the indications of a Human Centered Design (ISO 9241-210:2019) [57]. In addition, the proposed methodology explicitly integrated principles from User-Centered Game Design, incorporated aspects of the MDA framework (Mechanics, Dynamics, Aesthetics), and applied Playability Heuristics to ensure both usability and engagement. This combination provides a structured and replicable framework that connects therapeutic design criteria with engaging game mechanics, reinforcing the methodological rigor of the design process.

The mini-games were designed for mobile devices with the Android operating system and developed using Unity 3D version 2019.2.3 [52]. The 3D models used in the video games were created using Blender [58]. The gyroscope and accelerometer of the mobile device are utilized to provide various gameplay options. The AR reward game presented has been developed based on the design principles of learning games with AR for mobile devices, targeting children [23]. This game, created with AR visualization technology, utilized Google's ARCore platform [51].

The three mini games developed will be abbreviated as G1: "*Eliminate the virus*", G2: "*Fly the plane*" and G3: "*Fill the tank*" games. The common characteristics of these games are described below.

The start and finish screens of the games are similar, presenting the start screen the following options (Fig. 2):

Fig. 2. Example of "*Fly the plane*" start screen.

- **Practice:** a video tutorial showing the game mechanics is presented, illustrating how to move the mobile phone, the hands, and any action required to play. Subsequently, players will have the option to proceed to the practice mode, where they can spend time familiarizing themselves with the game mechanics before starting the actual gameplay.
- **Play:** provides access to play the games. Each game has three difficulty levels: easy (L1), medium (L2), and hard (L3), lasting 120 seconds per level.
- **Data:** shows a summary of the child's progress, including the game name, date of the game, difficulty level with the number of successful hits and failures of the presented stimuli, and whether or not the game was completed. This information is intended for therapists to obtain results.
- **English or Spanish:** allows access to the application in the selected language.
- **Exit:** closes the game.

In each game there is an end-of-level screen and an end-of-game screen. The first shows the level completed, the number of points earned, and a motivational phrase to positively encourage the child. The second one shows the current score earned in the game, the score earned in the previous game, and a "*Continue*" button, which will lead to the next screen, where the player is offered the option to play the reward game or return to the menu to continue the game. Some individual features and game mechanics were developed for each game:

Eliminate the virus game (G1) The game's virtual environment is set in a hangar. Within this hangar, there is an area containing a translucent pipe through which particles, resembling virus-like images, can be seen (Fig. 3). The objective of the game is to eliminate these particles using a laser ray.

Particles appear randomly within the pipe, with a preset number assigned to each difficulty level. To eliminate a particle, the player must align it within the crosshairs displayed on the screen by moving the mobile device. Once aligned, a laser ray activates, and the player needs to keep the mobile device pointed at the particle for a certain time. If the player clears the pipe of all particles within the 120-second limit, an equivalent number of particles reappear. Difficulty increases with more particles and longer alignment times: L1 requires removing 10 particles, L2 requires 14 particles, and L3 requires 21 particles.

Fly the plane game (G2) The game is set in a natural environment, aiming to provide tranquility and serenity. It primarily relies on the colors of nature, known for their calming effects on human beings. The player's objective is to collect the hoops that appear, using a virtual plane. This plane is controlled by rotating the mobile phone, which acts as a lever: tilting the device towards the player raises the plane, tilting it

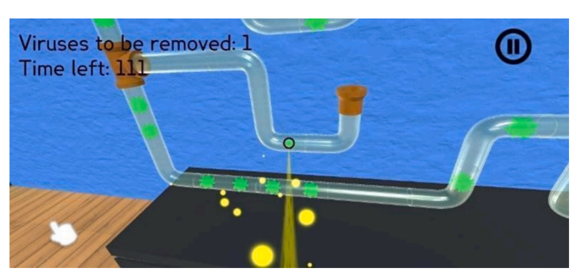

Fig. 3. Example of "*Eliminate virus*" screen.

away lowers it, and rotating it left or right moves the plane horizontally in the respective direction. The collision detection between the plane and the hoops is limited to the central body of the plane, excluding the wings. This ensures that players have to catch the hoops naturally without trying to cheat by touching the wings. In addition to collecting hoops, the natural environment includes mountains and water that act as obstacles. Collisions with these elements result in errors, requiring players to maintain control of the plane to avoid them. This feature was designed to enhance attention and visuomotor coordination by increasing the challenge of navigation. The difficulty levels are based on scene lighting conditions (Fig. 4): morning (L1), evening (L2), and night (L3). As lighting decreases, the player must be more attentive and focused to effectively identify and collect the hoops.

Fill the tank game (G3) The game takes place in a hangar environment, where the objective is to fill a carafe with particles emitted from a tap (Fig. 5, left) and then empty the carafe's contents into a tank (Fig. 5, right). Each time the carafe is positioned under the tap, the tap starts emitting particles, and stops only when the carafe is removed from the tap. Once the carafe is removed from the tap, the player must carefully carry it to a green tank located to the right of the tap without spilling any particles along the way. The challenge lies in maintaining control and coordination while transferring particles from the carafe to the tank.

To manipulate the carafe, players rotate their mobile device as if it were an actual carafe. Notably, the game requires players to stand and make slight movements, promoting visuomotor coordination. The game is also suitable for both left- and right-handed players. The difficulty increases as the tank's opening size decreases and transitions from being static to moving: tank with a large, static opening (L1), tank's opening is smaller than in L1 and remains static (L2), tank's opening is the same size as in L2, but the tank moves slightly (L3).

The reward game: Draw a graffiti At the end of each game, players are offered the chance to play a reward game, "*Draw a Graffiti*" This AR-based game was designed to promote creativity and self-expression, making the therapeutic intervention more entertaining and engaging. Children can create graffiti with their mobile device. The game allows movement and visual motor coordination by simulating a spray can. A virtual blackboard and four virtual spray cans—red, yellow, green, and blue— appear on the screen (Fig. 6, right), allowing players to paint freely on the virtual blackboard. Once the drawing is completed, the artwork is displayed on the screen and can also be printed as a motivational reward (Fig. 6, left).

This reward game is accessible to all players once they complete the three main mini-games, regardless of their performance or score. It is designed as a playful break to promote relaxation, self-expression, and motivation. The activity of drawing graffiti allows children to engage in creative, low-pressure expression while practicing visuomotor coordination through the spray-like interaction mechanics.

In order to clarify how feedback is delivered to children during gameplay, we included an interaction loop diagram (Fig. 7). This representation summarizes the therapeutic cycle in which each game stimulus prompts a child's response, immediately followed by system feedback. Such feedback consists of positive auditory cues, visual reinforcement messages (e.g., "Congratulations!", "Wow!"), and score updates, ensuring continuous motivation while avoiding frustration. This mechanism strengthens the therapeutic dimension of the games and guarantees consistency between the design principles and the expected outcomes. In addition, each game incorporates therapeutic checkpoints to ensure that progress is continuously monitored. These checkpoints consist of the registration of successes and failures during the game, the delivery of motivational reinforcement messages at the end of each level, the display of scores to provide immediate evaluation, and the option to access the reward game as a final motivational element. This structure guarantees that therapeutic objectives are embedded within the gameplay flow.

4. Prototype evaluation

The evaluation phase was designed to validate the proposed methodology in achieving its objectives: designing a game for children with ADHD. This phase integrated quantitative and qualitative methods to assess the games' playability, engagement, and adherence to the design criteria established during the analysis and design phases. In this phase, the two hypotheses set out in the introduction were tested: Hypothesis 1 evaluated the playability, engagement, usability, and accuracy of the games, addressing the Research Questions RQ1 to RQ5. Hypothesis 2: assessed the adaptation of the games to the specific needs of children with ADHD, corresponding to RQ6. This pilot study was designed to validate the playability, usability, and design compliance of the prototype games prior to clinical testing with children diagnosed with ADHD.

- **RQ1:** Was the playability of these video games well accepted?
- **RQ2:** Was the sympathy and enjoyment factor positive of these video games?
- **RQ3:** Was the ease of use dimension of these video games positive?
- **RQ4:** Was the quality factor of accuracy percentage of these video games high?
- **RQ5:** Was an experienced player more efficient than an inexperienced one in the number of hits scored in the games?

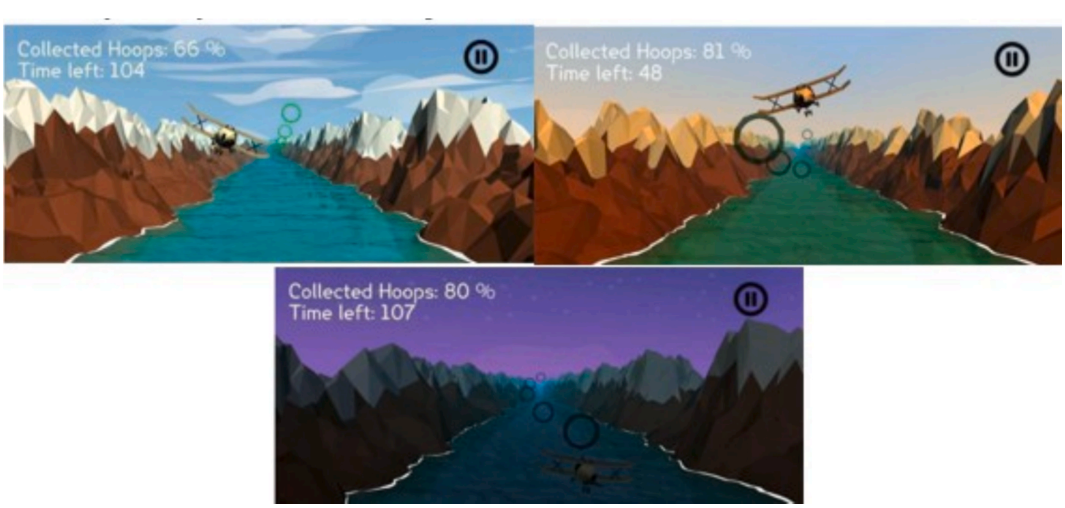

Fig. 4. Examples of the different level screens of the "*Fly the plane*" game. L1 (top left), L2 (top right) and L3 (bottom).

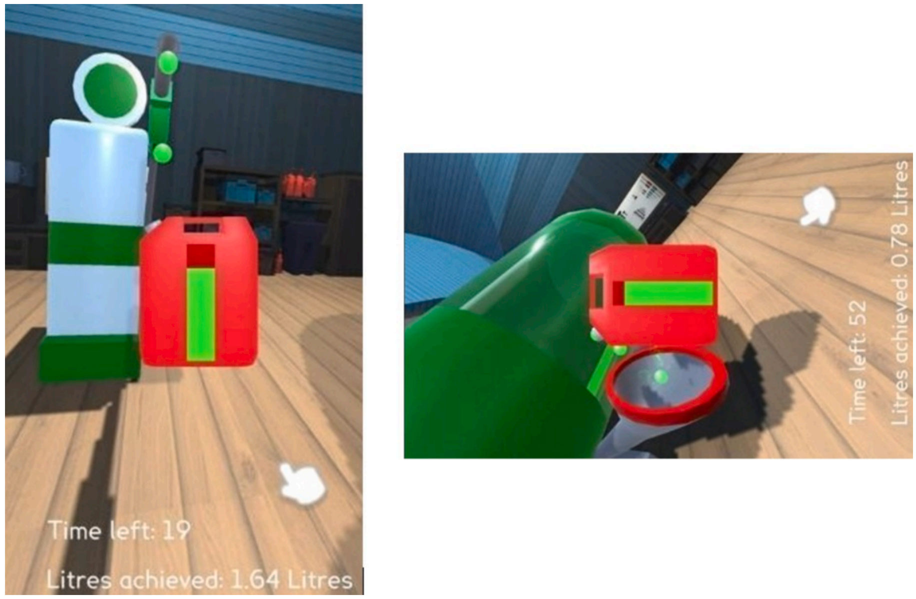

Fig. 5. Example scenes of “*Fill the tank*” game.

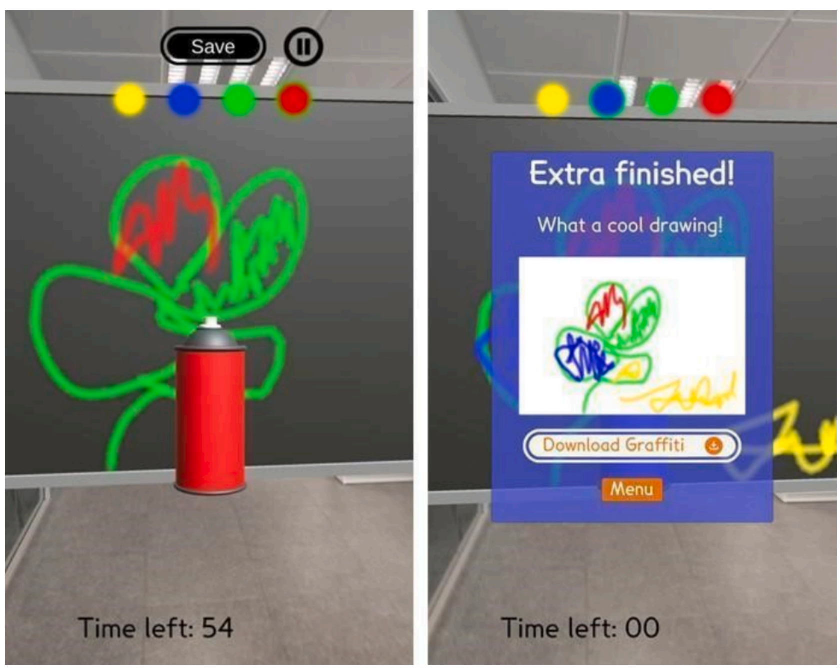

Fig. 6. Example scenes of “*Draw a graffiti* “AR game.

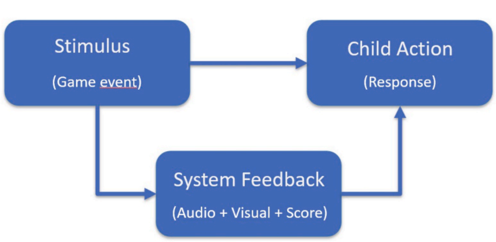

Fig. 7. Interaction Loop: Stimulus – Action – Feedback.

- **RQ6:** Were the key aspects included in the games to ensure the adaptation to the specific needs of children with ADHD well accepted?

These hypotheses guided the design of the evaluation process and the development of the instruments. A quantitative, cross-sectional methodology was applied to evaluate the quality attributes of the games in this exploratory study [59].

4.1. *Participants and design of the evaluation process*

The evaluation phase was not designed to measure the therapeutic effectiveness of the games on ADHD symptoms, but rather to validate playability, user engagement, and compliance with the design criteria previously established in the methodology. On the basis of these premises, the subjects involved in the test were neurotypical children and regarding the adult participants, three of them reported a formal diagnosis of ADHD; however, none of them were under pharmacological treatment at the time of the study. Once this preliminary validation was completed, a separate intervention phase with clinically diagnosed children with ADHD was initiated and will be presented in future studies. The study used two convenience samples, adults and children (Table 2). Standardized instruments such as the UES, SUS, or GEQ were not employed, as no validated versions were available for Spanish-speaking children aged 8 to 12. Instead, developmentally appropriate instruments were selected and adapted with accessible language, following expert recommendations and supported by Guerrero García and González Calleros [15], Vona et al. [60], González Sánchez and Gutiérrez Vela [61]. Questionnaires adapted for children were developed based on the playability questionnaire proposed by Vona et al. [60], González Sánchez and Gutiérrez Vela [61] and the PrEmo-type questionnaire used in Guerrero García and González Calleros [15]. All questionnaires were administered and completed in paper format during the laboratory sessions, under the supervision of the research team. Additionally, the criterion of judges was applied by four psychology students with 100% agreement. Inclusion criteria for the children are detailed in Table 2. Adult involvement, including parents and caregivers, was also key during the children's evaluation phase. Adults supervised the children throughout the study, performing the following tasks:

- Monitoring the children while they played.
- Ensuring the play area was free of distractions.
- Preparing a safe, obstacle-free space for standing play in G3.
- Sending the results to therapists for evaluation.

4.2. Instruments and research variables

Participants'age, gender, level of education and profession were retrieved via personal communication.

4.3. Sympathy & enjoyment and Ease of use

Following the playability model proposed by González Sánchez and Gutiérrez Vela [61] to evaluate the factors of playability and quality of the interactive experience of video games, an adaptation of the playability questionnaire by Vona et al. [60] was carried out, and some metrics of the attributes of effectiveness and efficiency were considered. Two questionnaires were developed, one for adults and one adapted for children. Each participant completed one questionnaire for each game tested. In the children's questionnaire each score was represented by a different color and face emoticon. Both questionnaires consisted of 9 items, evaluated with a 5-point Likert scale ranging from 1 (not at all) to 5 (very much). Although in this pilot study the response options were labeled from 0 to 4, this corresponds to a five-point structure, and future studies will adopt the conventional 1–5 format to ensure consistency in interpretation. The results were analyzed based on the summed total scores for each dimension, following the structure of the adapted instruments.

The items proposed for each questionnaire, grouped by the category they assess, are shown in Table 3. In addition, an open question was included in which the participants could make observations that have not been considered in the proposed items and that they considered relevant.

Gameplay questionnaire for adults:

- Sympathy and enjoyment ratings were obtained from the total score of items Q1, Q4, Q5, Q6 and Q8. The maximum score that could be obtained is 20.
- Ease of use ratings were obtained from the sum of the scores for items Q2, Q3, Q7 and Q9. The maximum score that could be obtained is 16.

Gameplay questionnaire for children:

- Sympathy and enjoyment ratings were obtained from the total score of items Q1, Q3, Q4, Q5, Q7 and Q8. The maximum score that could be obtained is 24.
- Ease of use ratings were obtained from the sum of the scores for items Q2, Q6 and Q9. The maximum score that could be obtained is 12.

In both playability questionnaires, the total score was obtained by summing the scores of all items, following the approach used in previous validated instruments. Higher scores indicated higher satisfaction, ease of use, and enjoyment (see Table 5), and the maximum total score was 36; therefore, the analyses and tables in this study report means of summed total scores rather than the average per item.

In addition, for both samples, the attributes of effectiveness and efficiency were considered as quality factors in video games [61]. To evaluate effectiveness, for each level we computed the accuracy

Table 2
Distribution of the participants in the experiment. Abbreviations: M (Mean); SD (Standard Deviation).

Description		Sample
Adults	Volunteers:	12 (6 males, 6 females)
	2 professionals in psychology field	20–40 years old
	2 professionals in video game design and development field	(M = 24.58; SD =5.83)
	5 adults without ADHD	
	3 adults diagnosed with ADHD (not under medication during the study)	
	- 1 ADHD-Inattentive Subtype	
	- 1 ADHD-Hyperactive Subtype	
	- 1 ADHD-Combined Subtype	
Children	Students from different public schools in Castellón (Spain)	13 (6 girls, 7 boys)
	Inclusion criteria:	8–12 years old
	(1) parents had given informed consent for their participation in the study	(M = 9.62; SD =2.43)
	(2) not diagnosed with any neurodevelopmental disorder	
	(3) IQ (intelligence quotient) indices were normal (reported by the family)	

Table 3
Gameplay questionnaire for adults and children.

Assessed categories	Adults' items based on the test for adults	Children's items based on the test for children
Sympathy	Q1. Degree of satisfaction with the video game	Q1. Did you like the video game?
and	Q4. Degree of involvement in the challenge of the game	Q3. Did you find it too long?
enjoyment	Q5. Degree of motivation for the game	Q4. Did you get involved?
	Q6. Degree of satisfaction with the reward system	Q5. Did you like the animation at the end?
	Q8. I would play more games with similar features	Q7. Would you like to play a similar game?
		Q8. How did you feel?
Ease of	Q2. Ease of understanding and mastering the	Q2. Was it easy to play?
use	mechanics of the video game	Q6.Did you understand the instructions?
	Q3. Adequate time to reach the objective of the game	Q9. Have you played more mobile games?
	Q7. Degree of understanding of the instruction menu	
	Q9. I am familiar with mobile games	

percentage (A/B*100, where A = number of hits and B = total stimuli).

To assess the degree of efficiency, familiarity of use was considered: How efficient was the expert user compared to an inexperienced player? For this purpose, participants were classified according to the score of item 9 of the questionnaires. Scores of 0, 1, 2 were classified as non-experts (5 adults and 3 children) and scores of 3 and 4 as experts (7 adults and 10 children). The scores of both groups were compared using the *Mann-Whitney U* test for accuracy percentage.

4.4. Adaptation of video games to the specific needs of children with ADHD

To demonstrate whether the key aspects included in the games ensured their adaptation to the specific needs of children with ADHD, a relationship was established between these aspects and the questions to be assessed in the adult and child questionnaires used in the experiment (Table 4). In addition, the maximum score that could be obtained in each of the key aspects assessed, was included in this table.

4.5. Procedure

The research was conducted in accordance with the 1964 Declaration of Helsinki and its subsequent amendments and with the approval of the Deontological Commission of the Jaume I University with file number “CD/83/2022”. Adult participants and parents of children who volunteered to participate received written informed consent together with an information sheet on the objectives and involvement in the research. Once written consent had been obtained from all participants, the video game evaluation session was carried out in a research laboratory with little noise pollution in the Faculty of Health Sciences at the Jaume I University. A one-hour session was used to play the three video games, dedicating approximately 8–9 minutes to each of them. Two to three minutes were dedicated to a preliminary phase of understanding and practice of the game, followed by the six-minute intervention phase. After playing each video game, the participants completed the playability questionnaire. All questionnaires were completed anonymously, and no identifying information was collected. Likewise, the data generated by the games were stored using numerical codes to ensure participant anonymity and data confidentiality.

5. Results

Due to the non-normal distribution of the data (confirmed by the *Shapiro–Wilk test*), our analyses were carried out using non-parametric statistics, with the Mann–Whitney U test for inter-group comparisons. Results were considered significant at $p < .05$, and are shown in Table 5. It should be noted that the values shown in Tables 5 and 6 represent the means of the total summed scores for each dimension, rather than the means of individual *Likert* items. This approach was adopted to maintain consistency with the original instruments used in previous studies. Given the exploratory nature of this pilot study and the limited sample size, the analyses presented in Tables 5 and 6 are descriptive only, providing preliminary insights rather than inferential evidence. Future studies with larger clinical samples will include formal statistical comparisons to assess significance. The results of this study are structured around two primary hypotheses. Hypothesis 1 encompasses RQ1, RQ2, RQ3, RQ4, and RQ5, and Hypothesis 2 addresses RQ6. The findings for each hypothesis are presented below.

Regarding Hypothesis 1, the playability, in both samples, G1 was the most highly rated, followed by G2 and G3. These results answered RQ1 affirmatively. The mean response score for the three games in the case of adults is approximately 2.7 and in children 3.04. Despite being higher for children than for adults, in both cases the playability of the games was considered acceptable because mean values above 2 indicate positive evaluations on the 5-point Likert scale. The analysis of the playability concept includes the concepts of sympathy and enjoyment of the game as well as its ease of use:

- Sympathy and enjoyment. G1 showed the highest degree of satisfaction and motivation for both adults and children, but this differed in the case of the other two games: adults were more satisfied with G3 than with G2, and the opposite was true for children. These results answered RQ2 positively. The mean response score for the three games in the case of adults was approximately 2.7 for adults and 2.94 for children. Despite being higher in the children than in the adults, in both cases the sympathy and enjoyment of the games were positive, as mean scores above 2 indicate favorable evaluations on the 5-point Likert scale.
- Ease of use. The results indicated that both adults and children found G1 the easiest to use, followed by G2 and finally G3. These results answered RQ3 positively. The mean response score for the three games in the case of adults was approximately 2.79 and in the case of children was 3.18. Although the score was higher for children than for adults, in both cases the mean value exceeded the threshold of 2, indicating that the ease of use of the games was acceptable.

Based on the quality metric of accuracy percentage, the results were segmented by each game level. These results are presented in Table 6. Upon analyzing the data, it was observed that:

- In G1, for both adults and children, the accuracy percentage decreased as the difficulty level increases (Fig. 8).
- In G2, the results were consistent across both groups, showing that accuracy percentage increased with difficulty level, with a slight decrease at the highest level (Fig. 9).
- In G3, while accuracy percentage decreases at the third difficulty level in both samples, the performance profiles differ; adults achieved higher accuracy at the second difficulty level compared to children (Fig. 10). These results provided a positive response to RQ4. It is noteworthy that, with the exception of G2 in the children’s sample, all games maintained accuracy levels exceeding 50% across all difficulty levels

To answer RQ5, the effectiveness of performance was analyzed based on the users’ familiarity with video games, comparing expert and non-expert players. The results in Table 7 show that there were no statistically significant differences between experts and non-experts in terms of success rate across the three games, both in adults and children. However, to complement the analysis, effect sizes were calculated using Hedges’s g to examine the magnitude of the differences between groups. In the adult sample, Hedges’s g indicated a moderate effect size for G1 ($g = 0.800$) and G2 ($g = 0.630$), and a small effect size for G3 ($g = 0.180$). In the child sample, the effect size was negligible for G1 ($g = 0.045$), moderate for G2 ($g = 0.778$), and large for G3 ($g = 1.597$). Despite these variations in effect size, the absence of statistically significant

Table 4
Key aspects assessed and questions from the gameplay questionnaire associated and maximum score that can be obtained.

Assessed key aspects	Adults’ items based on the test for adults	Children’s items based on the test for children
Motivational character	Q1; Q4; Q5; Q8. Max. score: 16	Q1; Q4; Q7; Q8. Max. score: 16
Playful activities	Q1; Q4; Q8. Max. score: 12	Q1; Q4; Q5; Q7. Max. score: 16
Without excessive auditory & visual stimuli	Q2; Q5. Max. score: 8	Q2; Q8. Max. score: 8
Degree of difficulty affordable	Q2; Q7. Max. score: 8	Q2; Q6. Max. score: 8
Estimated playing time	Q3; Q9. Max. score: 8	Q3; Q9. Max. score: 8
Immediate evaluation	Q4; Q8; Q5. Max. score: 12	Q4;Q8. Max. score: 8
Reward based system	Q6. Max. score: 4	Q5. Max. score: 4

Table 5
Results obtained from questionnaires related to playability, sympathy & enjoyment and, ease of use of the games. Abbreviations: M (Mean); SD (Standard Deviation).

	Game playability				Sympathy & enjoyment				Ease of use			
	Adults		Children		Adults		Children		Adults		Children	
Game	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
G1	26.50	4.96	29.15	4.07	14.83	2.95	18.76	3.05	11.67	3.02	10.38	1.55
G2	23.83	8.34	26.76	6.45	12.67	5.24	17.46	5.12	11.17	4.34	9.30	1.54
G3	23.66	7.81	25.84	7.86	13.00	4.30	16.84	5.11	10.67	4.25	9.00	3.31

Table 6
Results on the quality factor of accuracy percentage. Abbreviations: M (Mean); SD (Standard Deviation).

		Adults		Children	
Game	Level	M	SD	M	SD
G1	L1	84.83	20.26	85.08	9.68
	L2	64.25	14.03	55.31	7.80
	L3	59.00	10.05	51.15	6.55
G2	L1	58.25	20.46	38.46	21.31
	L2	71.58	16.97	42.46	19.38
	L3	69.33	20.18	42.31	21.31
G3	L1	81.33	14.46	72.46	23.75
	L2	83.67	7.63	66.00	21.69
	L3	77.92	9.17	67.54	19.49

differences suggests that prior expertise in video games did not systematically influence task performance. This finding supports the idea that the games were accessible and manageable regardless of previous gaming experience, which is a desirable characteristic in therapeutic or educational serious games aimed at diverse user profiles.

Finally, with respect the open-ended question proposed to participants, it is noteworthy that they reported a positive experience with the reward game, with children particularly expressing satisfaction in being able to take their drawings home as a reward of their participation.

Regarding Hypothesis 2, the acceptance of the key aspects incorporated in the video games, measured in both samples (adults and children) and answering RQ6, was overall positive, as shown in Table 8. It should be noted that the values in Table 8 correspond to the mean of the summed total scores for each key aspect rather than the average of the 5-point Likert items. This clarification ensures consistency with the scoring structure of the adapted questionnaires. In general, the highest scores were observed, for both adults and children, in the aspects of motivation, playful activities, and immediate evaluation, suggesting that the video games successfully captured attention and fostered participants' interest. This reinforced the idea that video games were effective in promoting user engagement and in- teraction. Regarding manageable difficulty and controlled stimulation, the scores reflected successful implementation. The games did not present excessive sensory stimuli and offered balanced challenges that participants perceived as appropriate for their abilities. However, areas for improvement were identified. Since the primary target users of the video games are children, it is particularly relevant to highlight that this group reported high satisfaction with the estimated playing time, indicating that the duration is well aligned with their attentional rhythm. Although some adults suggested minor adjustments in certain levels, this observation is considered secondary, as the final design validation is specifically oriented toward the pediatric population.

To enhance the interpretation of the findings related to RQ6, Fig. 11 presents a comprehensive visual representation of these results, distinguishing between children and adults. This figure depicts the means and standard deviations for the three games across each key aspect evaluated, offering a detailed comparison of the variability and central tendencies in the responses of both groups. The graphs revealed that children tended to achieve higher scores than adults in most aspects, particularly in Motivational Character and Playful Activities, indicating that these features held greater importance for younger participants. Conversely, adults exhibited higher average scores in areas such as Reward-Based Systems and Manageable Degree of Difficulty. The dispersion of responses varied across categories; for instance, aspects like Absence of Excessive Auditory and Visual Stimuli demonstrated high consistency, while others, such as Playful Activities, showed greater variability, particularly among children. Overall, children

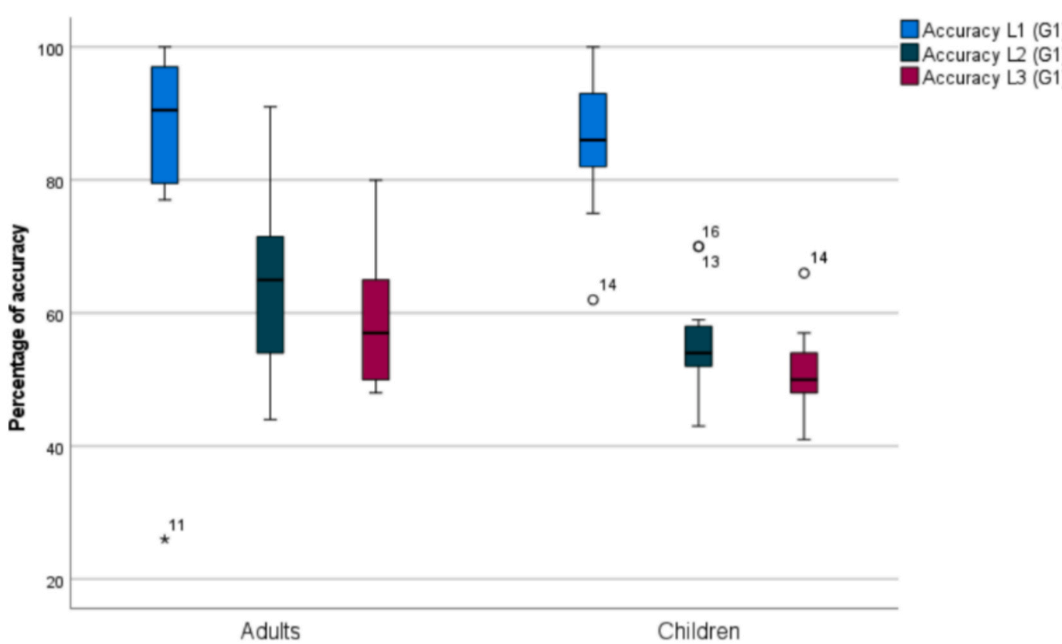

Fig. 8. Results of the quality factor of accuracy percentage of "*Eliminate the virus*" game. Easy Level (L1), Medium Level (L2) and Hard Level (L3).

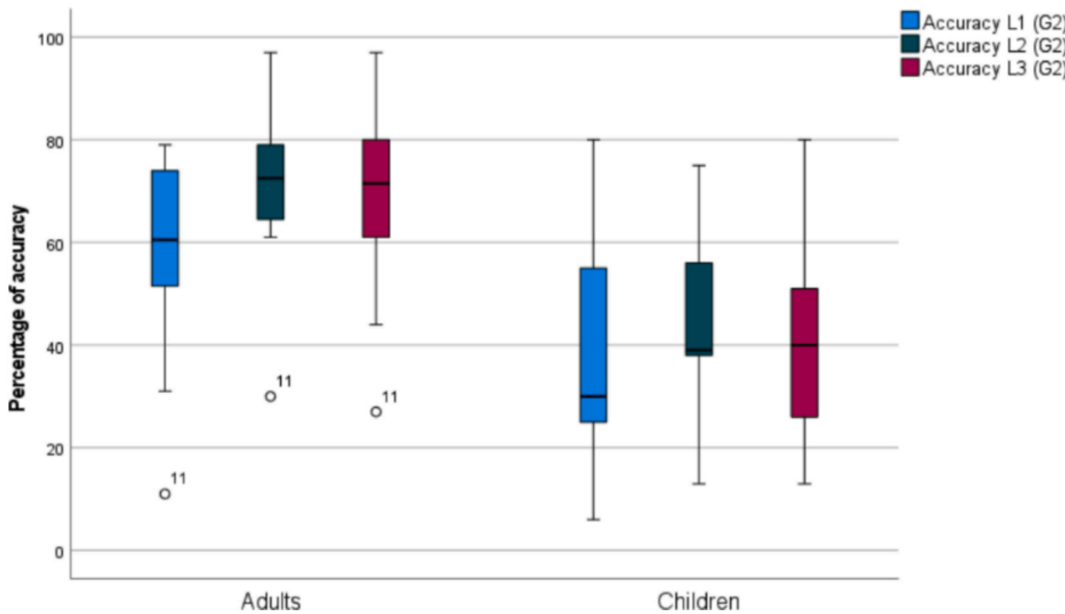

Fig. 9. Results of the quality factor of accuracy percentage of "*Fly the plane*" game. Easy Level (L1), Medium Level (L2) and Hard Level (L3).

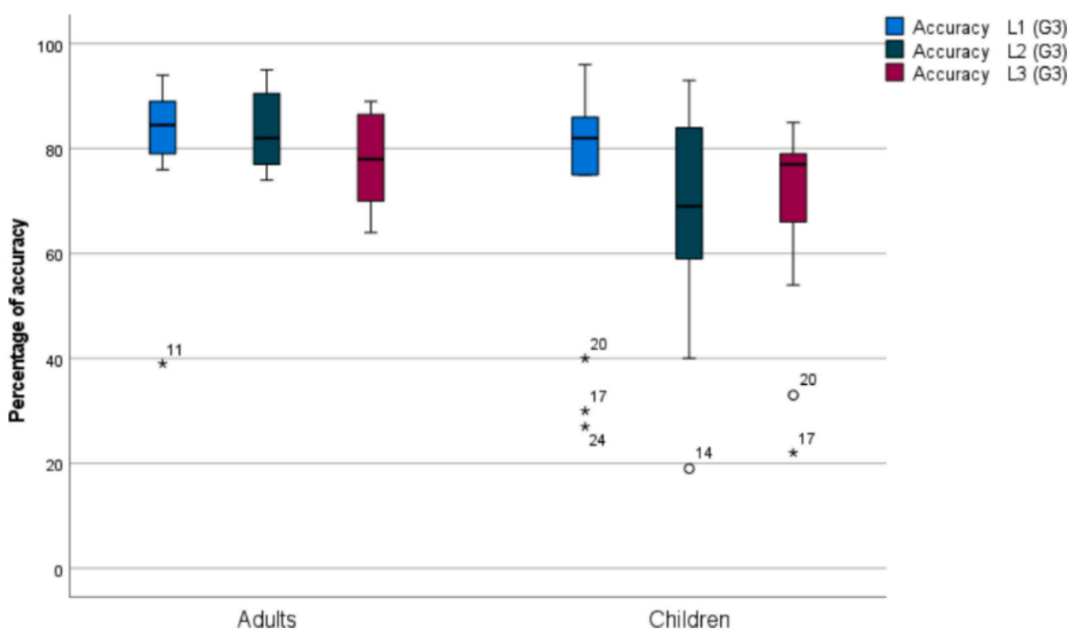

Fig. 10. Results of the quality factor of accuracy percentage of "*Fill the tank*" game. Easy Level (L1), Medium Level (L2) and Hard Level (L3).

Table 7
Degree of efficiency based on the comparison of user familiarity with video games.

Game	Adults *U Mann Whitney*	*p Value*	Hedges's g	Children *U Mann Whitney*	*p Value*	Hedges's g
G1	28	0.088	0.800	14.5	0.932	0.045
G2	23	0.372	0.630	7.5	0.204	0.778
G3	19	0.808	0.180	3.5	0.052	1.597

displayed more variability in their responses, suggesting a wider range of opinions on certain aspects, whereas adults exhibit greater consistency in their evaluations.

6. Discussion

At present, adherence to treatment in ADHD is still questionable. In fact, recently in the USA, the FDA [62] has prescribed the first treatment with video games for children between 8 and 12 years of age with ADHD as a parallel therapy for the improvement of the symptoms associated with the disorder. It is important to clarify that these serious games are not proposed as an alternative to pharmacological treatment, but rather as a complementary tool to support and reinforce therapeutic interventions in children with ADHD. Similarly, recent clinical evidence supports the feasibility of mobile-based serious games as complementary ADHD interventions [63]. This study presented a comprehensive methodology for designing and developing SGs tailored to children with ADHD.

This study involved a multidisciplinary team for the co-creation of

Table 8
Results for key aspect acceptance (means of summed scores). Abbreviations: M (Mean); SD (Standard Deviation).

Key aspects	Game	Adults M	Adults SD	Children M	Children SD
Motivational character	G1	12.33	2.23	13.23	2.35
	G2	10.25	4.18	11.92	4.01
	G3	10.50	3.63	11.54	5.13
Playful activities	G1	9.17	2.04	12.85	2.19
	G2	7.75	3.14	12.00	3.06
	G3	7.75	2.63	11.62	4.29
Without excessive auditory and visual stimuli	G1	6.42	1.00	6.69	1.03
	G2	5.25	2.09	4.92	2.16
	G3	5.50	2.02	5.46	2.44
Degree of difficulty affordable	G1	6.67	1.44	6.77	1.01
	G2	6.00	2.26	5.62	1.50
	G3	5.75	2.45	5.92	2.10
Estimated playing time	G1	5.00	2.09	6.15	1.99
	G2	5.17	2.44	6.15	1.81
	G3	4.92	2.02	5.31	1.89
Immediate evaluation	G1	6.50	1.17	6.69	1.25
	G2	5.08	2.31	6.15	2.23
	G3	5.75	2.14	6.00	2.45
Reward based system	G1	2.50	1.17	3.00	1.35
	G2	2.41	1.24	3.08	1.32
	G3	2.50	1.09	3.08	1.32

the video games formed by five professionals from three different academic areas: psychology, education and computer graphics. The pedagogical basis of the three video games has been informed by the literature and by experts in psychology in the care of vulnerable children and young students. The designers of the video games collected all the comments and, by consensus of the team, the games were developed. The findings demonstrated that a structured, evidence-based approach can result in tools that are both engaging and therapeutically effective. The success of this methodology lay in its emphasis on collaboration between multidisciplinary teams and its integration of user feedback at every stage of the process.

In order to assess the playability, quality and familiarity of use of the video games, two samples were included, one of adults with 12 subjects, 3 of them with a diagnosis of ADHD, and 13 children without any disorder.

The primary finding of this study is the alignment of scores between adults and children across the three video games in both playability and quality variables. Both groups rated satisfaction, ease of use, and enjoyment highly for each game, with scores ranging from 2.8 to 4. The highest-rated game was G1: E*liminate the virus*, followed by G2: *Fly the plane*, and lastly, G3: *Fill the tank*. Furthermore, one aspect to highlight from the results obtained was that the three video games can be applied to both expert and inexperienced subjects, as the user's familiarity with video games does not affect performance, which guaranteed similar chances of success for both expert and inexperienced players. The assessment of sympathy and enjoyment across both samples for the three video games indicated a high level of satisfaction and motivation, strong engagement in achieving objectives, and immediate positive performance feedback. Accordingly, and in line with the observations of Bashiri et al. [64], Avila-Pesantez et al. [65], De Luca et al. [47], these video games could serve as effective tools to support interventions for children with ADHD, as features like stimulus control, motivation, hand-eye coordination, reduced feedback time, and flexible usability may contribute to improvements in behavioral and cognitive skills.

In terms of ease of use, the results revealed a similar profile for both adults and children. Specifically, the video game G1: *Eliminate the virus* was the easiest to understand and master the mechanics of the video game. It also provided adequate time to reach the objectives and included a clear instruction menu. It is followed by the G2: *Fly the plane* with G3: *Fill the tank* being rated last. However, all three video games scored between satisfactory and highly satisfactory on user-friendliness. According to Zayeni and Revet [25], all these aspects (understanding of the mechanics of the video game, adequate time, and clear instructions) are essential qualities that contribute to effective educational game design for the studied child population.

Regarding accuracy levels, both in the video game G2: *Fly the plane* and in G3: *Fill the tank*, it was observed that as the level of difficulty increased, both samples improved their successful actions (e.g., collecting hoops or accurately transferring particles), indicating greater concentration and motor control. Interestingly, accuracy decreased at the third level of difficulty in all three video games, indicating that in all three video games the level of challenge increases. These findings aligned with prior research suggesting that video games can enhance concentration, processing speed, and visual discrimination. Including multiple levels of challenge helped balance the games, ensuring they are neither too easy nor overly difficult.

Analyzing the results by games and distinguishing between adults and children, it was established that, for children, G1: *Eliminate the virus* received the highest scores across most categories, particularly for motivation and playful activities, indicating strong engagement and a well-balanced difficulty-reward system. For adults, G1 also excelled, with high ratings for immediate evaluation and controlled sensory stimuli, highlighting its accessibility for both experienced and novice users. G2: *Fly the plane*, showed positive results, though slightly below G1. Children valued its playful activities and immediate evaluation but noted some repetitiveness in advanced levels, which may have affected sustained interest. Adults gave moderate ratings, appreciating its clarity and reward system but suggesting adjustments to difficulty for a more consistent challenge. G3: *Fill the tank*, received the lowest scores. Children found its initial difficulty somewhat high, causing frustration, though motivation and rewards were still positively rated. Adults also gave lower scores, citing excessive time to complete objectives, pointing to a need for shorter levels or adaptive difficulty options.

Finally, the evaluation of the key aspects incorporated into the video games, as perceived by both adults and children, yielded generally positive results. High scores were observed for motivational elements, playful activities, and immediate evaluation, indicating that these aspects effectively captured attention and sustained engagement. These findings aligned with Bashiri et al. [64], Avila-Pesantez et al. [65], highlighting the importance of motivational feedback and engaging gameplay in promoting user interest and adherence in therapeutic contexts. Regarding sensory control and difficulty levels, the findings confirmed that the games avoided excessive auditory and visual stimuli, offering an environment conducive to focus and concentration. The

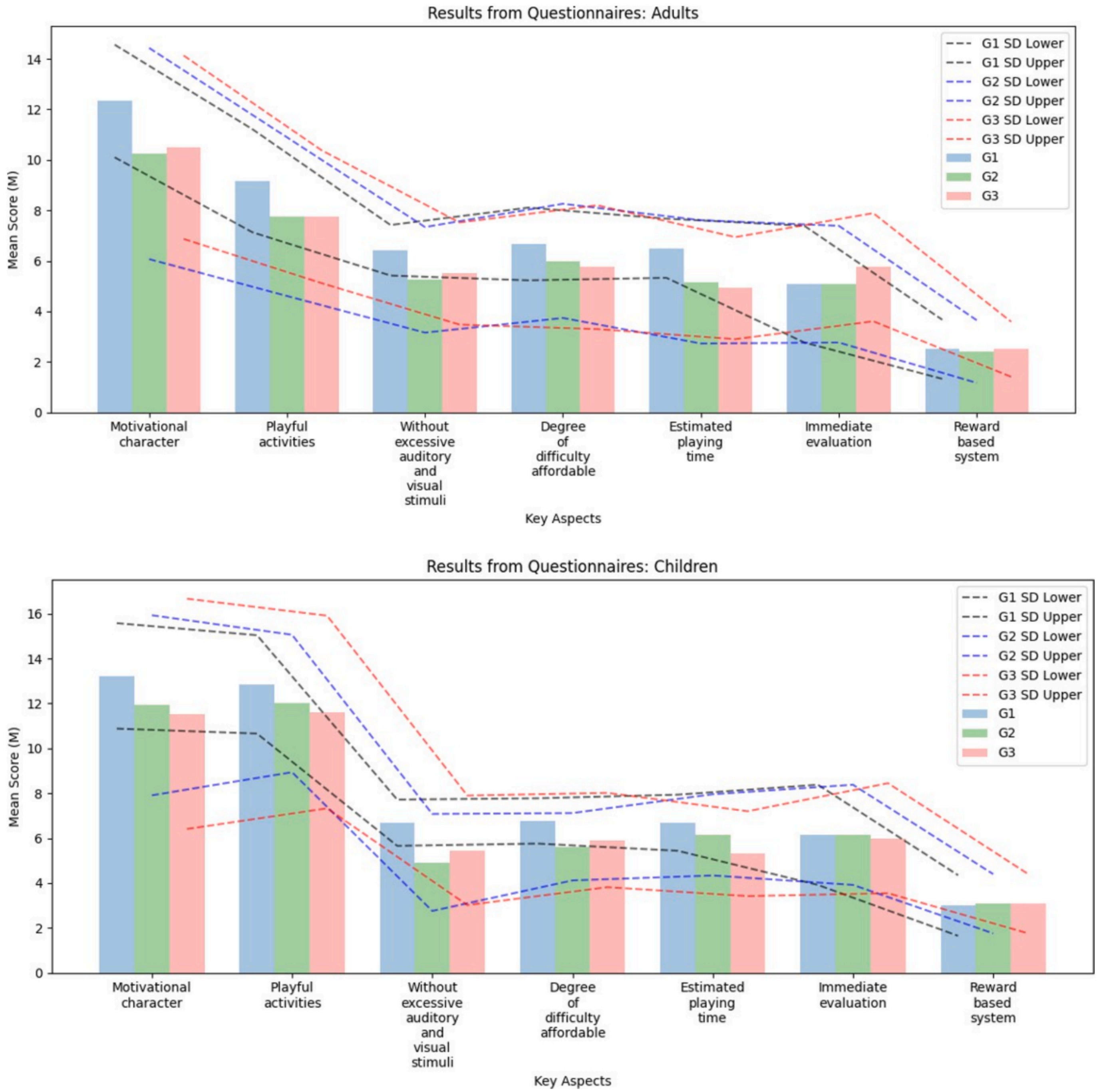

Fig. 11. Comparison of M and SD of the games in each of the key aspects evaluated: (top: adults and bottom: children).

difficulty levels were perceived as manageable, balancing challenge and accessibility to maintain user interest across different skill levels. This supported the idea that appropriately calibrated difficulty is critical for ensuring usability and engagement in educational and therapeutic games [66].

The analysis of means and standard deviations (Fig. 11) provided valuable insights into how the games addressed the distinct needs of adults and children. The observed differences, such as children valuing Motivational Character and Playful Activities more highly, and adults prioritizing Reward-Based Systems and Manageable Difficulty, underlined the adaptability of the game design to varied user profiles. These results suggested that incorporating diverse design elements ensured broader accessibility and relevance. The consistent ratings for Absence of Excessive Auditory and Visual Stimuli across both groups validated the balanced sensory environment of the games, while the variability in Playful Activities highlighted opportunities to refine engagement strategies for children. This reinforced the importance of tailoring game features to maintain attention and foster motivation across different age groups and user preferences. The findings also confirmed that the games are positively evaluated in terms of playability, sympathy, and enjoyment, and successfully integrate key design elements tailored to the needs of children with ADHD. These results validate Hypothesis 1, demonstrating that the games exhibit high levels of playability, engagement, usability, and accuracy across expert and non-expert users (RQ1–RQ5). Similarly, Hypothesis 2 was supported, as the results confirmed that the games successfully integrate key design elements, such as motivational aspects and manageable difficulty, ensured their adaptation to the specific needs of children with ADHD (RQ6).

Overall, these findings validate the proposed methodological framework, confirming its potential as a foundation for future clinical applications and larger-scale validation studies.

7. Limitations and future work

Nevertheless, these findings should be considered preliminary. The study involved a small pilot sample that included neurotypical children and three adults formally diagnosed with ADHD (not under pharmacological treatment). While the participation of these adults provided useful insights, the results from children cannot yet be generalized to clinical populations. Larger-scale studies with children diagnosed with ADHD are required to validate the therapeutic effectiveness of the proposed games and confirm the generalizability of the methodology.

The serious games developed in this study are designed to be integrated into real-world therapeutic settings, including schools, clinics, and home-based interventions under adult supervision. Their structure and mechanics allow therapists or caregivers to tailor the experience to each child's cognitive profile. Each game session automatically records user performance data (e.g., number and type of errors, interaction patterns, and completion time), which can be used to track progress, adapt therapy, and support decision-making.

In future research, we plan to explore the potential for both primary use of this data in therapeutic follow-up and secondary use in scientific research or regulatory applications. This includes examining how in-game performance metrics might complement traditional assessment tools, enhance individualized interventions, or contribute to evidence in support of digital health solutions. Although no direct transfer tests were conducted in this pilot phase, the design was informed by previous evidence indicating that serious games targeting attention, inhibitory control, and visuomotor coordination can foster improvements in related real-world tasks, such as those reported by De Luca et al. [47].

Furthermore, we acknowledge the importance of comparing serious games with traditional interventions such as occupational therapy to provide a more comprehensive perspective on ADHD treatment options. Occupational therapy is a more traditional, personalized intervention typically conducted with the help of an occupational therapist in a physical setting, focusing on manual and sensory activities. In contrast, serious games offer a more accessible and flexible alternative, allowing children to practice skills such as attention, inhibitory control, and motor coordination in a more interactive and engaging format. The integration of technology in serious games offers advantages such as customization, remote accessibility, and potentially greater adherence to treatment protocols, especially in home or school settings.

There are certain limitations in this study that should be addressed in future research. Firstly, the collected data should include not only the number of errors but also the type of errors, specifically distinguishing between accuracy errors, errors due to lack of control, and errors resulting from inactivity. For example, in the G2: *Fly the plane*, it could be examined whether children have missed the hoops (accuracy error), if children have crashed into mountains or water (error due to lack of control) or errors due to passivity (inactivity error). Secondly, although the degree of accuracy decreases in all three games, the repetition of similar challenges (three levels) could feel boring. However, as Laine and Lindberg [66] point out, if the game aims to develop a skill, repeatable challenges can be beneficial because they relate to control, as long as the player can choose the number of repetitions. Thirdly, it would be interesting to extend the sample of children to other subjects with disorders, e.g. with specific learning disabilities. Previous research indicates that some disorders may benefit from potential effects on variables such as attention and motivation. Fourthly, a usability requirement of G3: *Fill the tank* involves physical movement, so the use of familiar control, a safe and comfortable space is necessary. Finally, further fine-tuning of the timing is necessary to maintain interest without causing fatigue or demotivation.

Regarding the methodology used, future research should focus on refining this framework, particularly by exploring additional feedback mechanisms and expanding its application to other neurodevelopmental disorders. By documenting this methodological approach, we aimed to contribute to the broader field of therapeutic game development and provide a model for creating accessible, effective tools for diverse populations.

In this context, future work will explore advanced data analysis techniques and the integration of adaptive difficulty mechanisms to enhance personalization and sustain motivation during therapeutic gameplay. These approaches may include Artificial Neural Networks (ANN), log-linear logic mining frameworks, and evolutionary algorithms, which can reveal behavioral and therapeutic patterns that traditional statistical methods might miss. In addition, future research will extend the evaluation to children clinically diagnosed with ADHD to validate the therapeutic impact of the games in real-world settings.

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CRediT authorship contribution statement

Cristina Rebollo: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Rosa García-Castellar:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Inmaculada Remolar:** Writing – review & editing, Supervision, Investigation. **Veronica Rossano:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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